

Nameer Iqbal Ansari

RTL Engineer | FPGA | RISC-V

✉ nameeriqbalansari@gmail.com  [linkedin.com/in/nameer-ansari](https://www.linkedin.com/in/nameer-ansari)  github.com/Nameer-Iqbal-Ansari  [Visit My Website](#)

☎ +92 3363033155 📍 Karachi, Pakistan

PROFESSIONAL SUMMARY

RTL Design Engineer (FPGA/ASIC) at XCLR Technologies, specializing in **RISC-V-based SoC** development. Certified **RISC-V Foundational Associate (RVFA)** and **Google Summer of Code (GSoc) 2022 contributor** and **2023 mentor**, with hands-on experience in **SystemVerilog/Verilog** based **ASIC** and **FPGA** designs, **Lint/CDC/RDC** sign-off, **timing closure**, and **verification**.

WORK EXPERIENCE

02/2025 – Present

RTL Design Engineer, XCLR Technologies (Pvt.) Ltd.

- **AWS EC2 F1 Shell-Compatible SoC Subsystem** — Developed and validated a subsystem making the SoC's **JTAG**, **UART**, and **DDR4** interfaces compatible with AWS EC2 F1 Shell's memory-mapped interface (**OCL/BAR1**) and **DDR** controller. Verified interface accessibility through **C++** host-side drivers across both F1 emulation and AWS EC2 simulation environments
- **VU19P FPGA SoC Bring-Up** — Led development and emulation of an SoC targeting the VU19P FPGA using **Xilinx Vivado** for **synthesis**, implementation, and **bitstream** generation. Integrated **UART**, **DDR**, and **JTAG** interfaces, and validated **DDR** and **JTAG** functionality through on-hardware emulation testing on the VU19P board
- **PULP BSCANE2 TAP Integration for Unified JTAG Access** — Integrated PULP's **dmi_bscane_tap** into the design's functional JTAG infrastructure, enabling a single **Xilinx HS-3 cable** to serve dual purposes through the FPGA's native JTAG pins (defined in the **XDC**) bitstream configuration via the FPGA's own TAP controller, and functional **RISC-V debug** access via the **BSCANE2** primitive, which internally taps the user-defined scan chain to expose PULP's debug module as the design's virtual TAP.
- **DDR4 Bring-Up & Linux Boot on AWS EC2 F1 FPGA** — Interfaced the SoC with AWS EC2 F1's **DDR4-C** shell controller to bring up external memory, then loaded the Linux boot image into **DDR** and validated end-to-end boot on the Ariane (**CVA6**) RISC-V core, confirming memory-mapped access and core-to-DDR data integrity through the shell interface
- **JTAG Bring-Up & Debug Validation on AWS EC2 F1** — Bridged the design's JTAG interface to AWS EC2 F1 shell's JTAG connectivity, with **OpenOCD** running as the intermediate server translating RISC-V **GDB's** remote protocol (host-side) into JTAG scan commands driving the design's debug module (target-side) on the F1 emulation instance. Ran full test cycles covering core halt/resume, **single-stepping**, **breakpoints**, and **register/memory access**, validating consistent debug behavior across repeated test runs

09/2023 – 02/2025

Associate Engineer, Xcelerium

- **AWS EC2 F1 Shell Interface Validation** — Verified 4 AWS EC2 F1 shell interfaces (**OCL**, **BAR1**, **DDR**, **SDA**) using **C++ drivers** and SystemVerilog testbenches, across both AWS EC2 F1 emulation and simulation environments.
- **Logic-Analyzer-Based Debugging** — Debugged JTAG and SoC subsystem behavior using logic analyzers across both AWS Cloud emulation and VU19P FPGA hardware environments
- **Technical Documentation & Design Reviews** — Authored comprehensive design documents, verification test plans, timing diagrams, and block diagrams to aid design reviews and knowledge transfer

03/2020 – 09/2023

Research Assistant, Micro Electronics Research Lab (MERL)

- **RV32I Core Design** — Built an open-source **RV32I CPU core**  from scratch in SystemVerilog, simulated with **Verilator** and **FuseSoC**, and emulated on an **Arty A7 FPGA** with **Xilinx Vivado**. Later integrated basic **UVM** and **JasperGold** support
- **Out-of-Order Upgrade to RV32I Core** — Extending the **RV32I** core with out-of-order execution capability, adding 2-wide instruction fetch, register renaming via a **Register Alias Table (RAT)**, a reorder buffer (**ROB**) for in-order commit, and reservation stations feeding ALU and branch execution units. Migrating the existing in-order pipeline to a **dispatch/issue/commit** structure inspired by **Berkeley's BOOM** design
- **NOVA1 SoC Development** — Contributed to **NOVA1 System-on-Chip (SoC)** development and its **AWS EC2-based** emulation, and **Zyper OS** bring-up
- **Multicore Cache Coherency & CHI Research** — Led the research of the **multicore architecture** and its **cache coherency** through the **OpenPiton** framework. Also researched converting OpenPiton's native coherence protocol to ARM's **CHI**, mapping its directory-based coherence engine onto CHI's **RN-F/HN-F/SN-F** node abstractions
- **RISC-V Training & Mentorship** — Conducted **student training programs** on RTL design and how to contribute to open-source
- **NOVA1 Presentation** — Presented NOVA1 project at the **RISC-V Karachi Meetup**

TECHNICAL SKILLS

- **RTL Design** — RISC-V VLSU | JTAG | SerDes | PCIe | eFuse | UART | CDC | XBAR | Cache | AXI | APB | TileLink
- **EDA Tools** — VCS | Verdi | Xilinx Vivado | GTKWave | CoreAssembler | CoreConsultant | QuestaSim | Verilator | FusionCompiler | PrimeTime
- **Formal Verification** — JasperGold | SVA (SystemVerilog Assertions)
- **Static Analysis** — Synopsys SpyGlass (Lint, CDC, RDC)
- **Languages** — Verilog | SystemVerilog | SystemVerilog-OOP | UVM (basic) | C | C++ | RISC-V Assembly
- **Scripting** — Tcl | Shell | Makefile | CI/CD | Python | YAML (FuseSoC)
- **Version Control** — GitHub | GitLab
- **Operating System** — Linux | Windows

SOFT SKILLS

Languages — English (C1), Urdu (Native)

Other — Design review | Technical Documentation | Debugging & Root-Cause Analysis | Design verification coordination | Technical Leadership | Timing-Closure Problem Solving | Research & Mentorship | Volunteer work

PROJECTS

Google Summer of Code (GSoC), Contributor (2022) / Mentor (2023)

- Added **FuseSoC** support on the **Azadi-SoC** and **rebased** the **lbex core** to latest upstream commits ([GSoC'22](#))
- Mentored **TileLink** Uncached Heavyweight (**TL-UH**) implementation in **Azadi-SoC** ([GSoC'23](#))

MARCore & NOVA1, Research Lead / Contributor

- Researched multicore cache-coherent memory systems using OpenPiton workflow and Ariane Core ([MARCore1](#))
- Contributed to NOVA1 SoC that contains RV64GC Core, UART, BRAM, DDR, and JTAG support ([The NOVA Project](#))

EDUCATION

2019 – 2023

Bachelor of Engineering in Electrical Engineering, Usman Institute of Technology

(CGPA: 3.289) (Specialization: Computer Systems)

- **Major Courses** — Digital Logic Design | Computer Architecture | Microprocessor-Based Systems | Embedded Systems | Operating Systems

Honors & Awards

- Hasham Foundation Scholarships by Usman Institute of Technology in 2022
- Linux Foundation (LiFT) Scholarship Recipient in 2024

CERTIFICATES

RISC-V Foundational Associate (RVFA) — Date: Aug 2025 | Issuer: RISC-V International | Sponsor: Linux Foundation | [Link](#)

RISC-V Fundamental (LFD210) — Date: Aug 2025 | Issuer & Sponsor: Linux Foundation | [Link](#)

PUBLICATIONS & ARTICLES

- Multicore SoC For SMP Linux ([Thesis](#))
- NOVA Emulation Project- An open-source RISC-V-based application-class AI SoC ([Project Doc](#))
- A Summer of Opportunity ([Medium — GSoC experience](#))
- Abstractions Can Lead Humanity to Ignorance (An Unpopular Opinion) ([Medium — technical opinion](#))